

Bhargi Bhalara

bhargibhalara03@gmail.com | 9735806664 | San Jose, California | linkedin.com/in/bhargibhalara

SUMMARY

Dynamic hardware engineer with over eight years of experience specializing in FPGA integration within complex systems. Proven expertise in RTL design for SoC/FPGAs/ASIC, VLSI, digital and analog circuit design, and embedded systems. Demonstrated ability to enhance system architecture through meticulous documentation and automation, significantly improving efficiency and accuracy in high-speed digital designs.

EXPERIENCE

Electrical Systems Engineer

Reflexion Medical, Inc.

Jun 2021 - May 2025

- Developed a comprehensive end-to-end FPGA design flow, including Synthesis, Implementation, and Bitstream Generation for Xilinx Ultrascale FPGAs using the Vivado design suite, enhancing design efficiency by 30%.
- Optimized existing state machine architectures, successfully reducing operational delays by 3-4 clock cycles, leading to a 15% increase in system performance.
- Automated regression testing processes using Python, which improved test coverage by 40% and reduced manual testing time by 50%.
- Created embedded test programs and validation protocols for bare metal testing, achieving a 95% pass rate during initial validation phases.
- Debugged high-speed serial interfaces, including PCIe and CAN Bus, utilizing protocol analyzers to ensure signal integrity, which resulted in a 20% reduction in signal errors.
- Initiated engineering change orders and facilitated the design transfer process for updated FPGA designs, ensuring compliance with regulatory standards and enhancing product reliability.

Electrical Engineer

Mirion Technologies (Capintec) Inc.

Apr 2017 - Jun 2021

- Redesigned RTL logic for FPGA/CPLD used in Multi-Channel Analyzers, improving processing speed by 25% and enhancing overall system functionality.
- Supported design changes for critical electronic components including Microcontrollers and SDRAM, ensuring compliance with ISO-13485 and IEC-60601 standards, which improved product reliability and market readiness.
- Delivered design changes to production with a 100% on-time delivery rate, adhering to strict design control processes and minimizing production downtime.
- Led hardware validation for Dose Calibrators and Radiation Measurement Devices, achieving an accuracy rate of 99.9% in measurements, which significantly boosted customer satisfaction.

Electronics Engineer

Reliance Electronics Inc.

Jan 2017 - Sep 2017

- Designed and validated printed circuit boards (PCBs) using custom test fixtures, ensuring correct functionality post-manufacture, which reduced error rates by 30% during initial testing phases.
- Structured and maintained the Bill of Materials (BOM) for circuit designs, effectively managing part selection and updates due to EOL/obsolescence, which minimized production delays by 20%.

Design Engineer

Scalable Systems Research Labs

Feb 2016 - Dec 2016

- Developed an IEEE Floating Point Numbers Multiplier in Verilog for the Floating Point Unit used in RISC-V CPU, enhancing computational speed by 15% through the implementation of advanced architectural techniques.
- Implemented a vertical slice compression technique to reduce carry propagation delays, resulting in a 20% increase in processor speed and efficiency.

FPGA Intern

Finisar Corporation

Jun 2016 - Sep 2016

- Engineered comprehensive Verilog code for 10G Ethernet transceivers on Xilinx Kintex 7 FPGA, optimizing data throughput and power efficiency, leading to a 10% reduction in power consumption during operation.
- Enhanced state machine design for Microsemi FPGA, improving operational efficiency by 15% and ensuring compliance with industry standards.

Instructional Student Assistant

San Jose State University

Aug 2014 - Dec 2014

- Assisted students in Advanced Digital Logic courses, enhancing their understanding of digital design concepts, which resulted in a 25% increase in student performance on assessments.

EDUCATION

Masters of Science in Electrical Engineering

San Jose State University, San Jose

Dec 2015

GPA: 3.4

Bachelor of Engineering

Sarvajanik College of Engineering and Technology

May 2013

SKILLS

AMBA (AHB, APB, ASB, AXI), ASIC, C, C++, Chipscope/ModelSim/Questasim, JIRA, JTAG, Lab Equipment and Tools, Linux, Mentor Graphics PADS, Shell, SOC, System Verilog, USB, Verilog, CPU Architecture, GIT, HDMI, Intel Quartus II/Prime, Jenkins, Microchip Libero SoC, PCIe/PCI, Python, QNX, QuestaSim, RTOS, VHDL, Vitis HLS, VUnit verification framework, Xilinx ISE Design Suite, Xilinx Vivado Design Suite